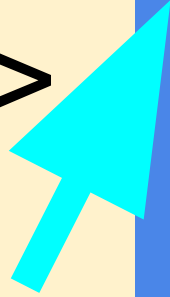


ONE PERSON IN GROUP-> CLICK USE TEMPLATE



WHEN THE COPIED TEMPLATE OF THESE SLIDES OPENS IN A NEW TAB:

- ADD YOUR Group NAME TO THE TITLE
- SHARE WITH GROUP MEMBERS (USE SHARE BUTTON)
- COPY THE TITLE AND LINK TO YOUR RECORDS (Digital folder/table of contents)
- Remember it is advisable to make a PERSONAL copy and link it when DONE

Concentration & Molarity

ESSENTIAL QUESTIONS

Remember, you should be able to thoroughly answer these AFTER learning through following resources.

- How are concentrations calculated?
- What are the different ways that concentration can be represented (m/m,v/v,m/v)?
- What are the different ways that very small concentrations can be represented (ppm, ppb)?
- How are concentrations for reactions represented (M, mol/L)?

Instructions

Follow through the **Engage to Apply** resources in order .

- Once complete, choose from Extend resources as interested.
- Record your learning in your **these slides- Remember to keep track of where to find these slides for studying!**
- Complete the **Essential Questions Response in the Apply Section** .
- Try the Extend section and optional Apply practice

Hyperdoc Elements

Click for quick access to:

- [Engage](#)
- [Explore](#)
- [Explain](#)
- [Apply](#)
- [Extend](#)

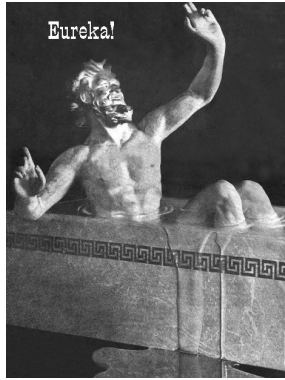
Terminology - As you encounter terms new to you, define in your own words here, include examples (make them up!)



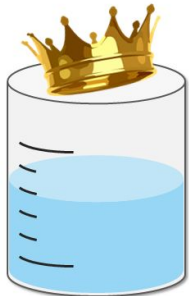
Part 1: ENGAGE

Explore - Part A: 5 minutes

There is a famous story with Archimedes:



A local tyrant, Hiero, contracts the ancient Greek polymath Archimedes to detect fraud in the manufacture of a golden crown because he suspects his goldsmith replaced some of the gold in his crown and replaced it with silver. Archimedes accepts the challenge and, during a subsequent trip to the public baths, realizes that the more his body sinks into the water, the more water is displaced--making the displaced water an exact measure of his volume. Because gold weighs more than silver, he reasons that a crown mixed with silver would have to be bulkier to reach the same weight as one composed only of gold; therefore it would displace more water than its pure gold counterpart. Realizing he has hit upon a solution, he leapt out of the bath and rushed home naked crying "Eureka! Eureka!" Or, translated: "I've found it! I've found it!"



Discuss with your group: How might the concept of moles be applied to the scenario of 'how much' of something there is in a given volume? In what other parts of your life is idea of the 'amount of stuff' in a given volume relevant? ***Make a list!***

Explore - Part B: 5 minutes

Option 1

Moles of solute
per Volume
(liters of solution)

vs

Option 2

Moles of solute
per Weight
(kilograms of solvent)

In your groups, discuss the advantages and disadvantages for representing concentrations with the two ways above. (hint: think about accuracy, usability, the impacts of temperature [like we learned with solubility], impacts of pressure, solute state of matter etc)

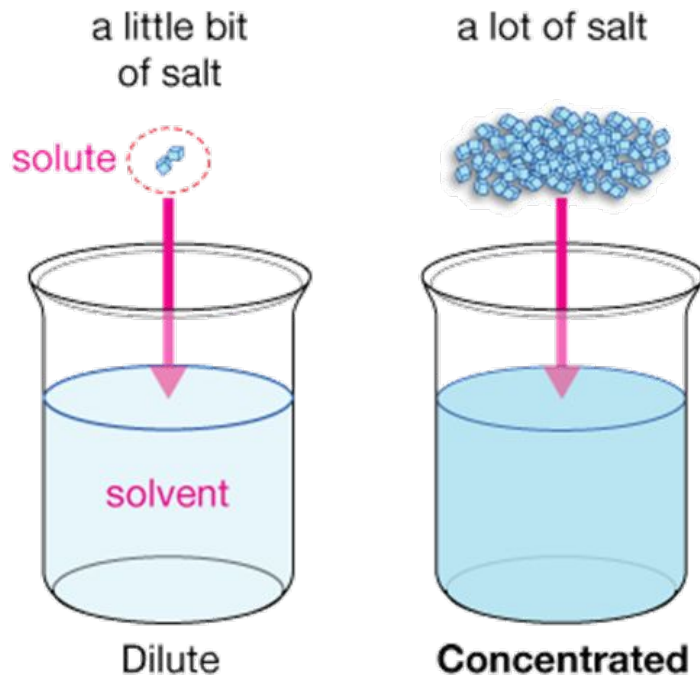
Give as many ideas as you can, correct or not! (don't self edit!)

Part 2: EXPLORE

Explore - What is concentration?

One way to describe solutions (qualitatively) is with the following terms:

- **Concentrated:** *high ratio of solute to solvent*
 - *ie. lots of solute in the solvent*
- **Dilute:** *low ratio of solute to solvent*
 - *ie. small amount of solute in the solvent*

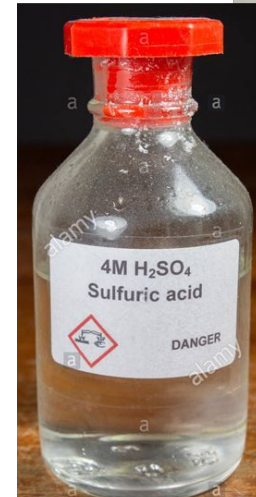


Explore - Quantifying Concentration

We can also quantify (put a number to) Concentration

This can take many forms:

- Percent concentrations (v/v, m/v, m/m)
- Parts per mass (ppm, ppb)
- Molar concentration (mol/L or M)
 - Most used in science



Thành phần khoáng chất (mg/l)	
Mineral Contents (mg/l)	
Na ⁺ :	45
K ⁺ :	8
Ca ²⁺ :	160
Mg ²⁺ :	14
Fe ³⁺ , Fe ²⁺ :	0.05
I ⁻ :	0
HCO ₃ ⁻ :	232
F ⁻ :	1.0-1.4
NO ₃ ⁻ :	0.2
SO ₄ ²⁻ :	400
H ₂ SiO ₃ :	70

TDS	
Total Dissolved Solids	
Ideal	0-170 ppm
Recommended	<500 ppm
Unacceptable	>1000 ppm

Source: Canadian Drinking Water Guidelines
© Wahl Water. All Rights Reserved

Explore - Percent Concentration

Percent Concentrations - ratio of **solute** to **solvent**, expressed as a percentage.

1. Mass Percent (m/m%)

➤ ratio of the mass of the **solute** to the mass of the **solution**

2. Volume Percent (v/v%)

➤ ratio of the volume of the **solute** to the volume of the **solution**

3. Mass/Volume Percent (m/v%)

➤ ratio of the mass of the **solute** to the volume of the **solution**

The choice of which one is based on the units (mass or volume) you have!

Explore - Calculating Mass Percent

$$\text{percent (m/m)} = \frac{\text{mass of } \textbf{solute}}{\text{mass of } \textbf{solution}} \times 100\%$$

****Units must be same for both****

Example: A 6.45g Sterling Silver ring contains 5.97g of pure Silver. Calculate the Mass Percent of pure Silver in the ring.

$$\text{Percent (mm)} = \frac{5.97g \text{ Pure Silver}}{6.45g \text{ Sterling Silver}} \times 100\% = 92.5\%$$

Explore - Calculating Volume Percent

$$\text{percent (v/v)} = \frac{\text{volume of } \textbf{solute}}{\text{volume of } \textbf{solution}} \times 100\%$$

****Units have to be the same****

Example: Acetic Acid is a liquid at room temperature. How much water (solvent) should be added (in mL) to 15.0mL of Acetic Acid produce a 5.00% (v/v) solution?

$$\text{volume of } \textbf{solution} = \frac{\text{volume of } \textbf{solute}}{\text{percent (v/v)}} = \frac{15.0\text{mL}}{0.0500} = 300\text{mL}$$

$$\text{Vol}_{\text{H}_2\text{O}} = \text{Vol}_{\text{Sol}} - \text{Vol}_{\text{Acetic}} = 300\text{mL} - 15\text{mL} = 285\text{mL}$$

Explore - Calculating Percent Mass by Volume

$$\text{percent (m/v)} = \frac{\text{mass of } \textbf{solute}}{\text{volume of } \textbf{solution}} \times 100\%$$

****Units of grams and mL needed****

Example: A solution was prepared by dissolving 17.5g of Glucose in water to make a 350mL solution. Calculate the percent (m/v) to 3 sig figs.

$$\text{percent (m/v)} = \frac{17.5\text{g Glucose}}{350\text{mL Solution}} \times 100\% = 5.00\%$$

Explore - Measuring Small Concentrations

Sometimes even very small concentrations are important to know.
(such as 0.0000001% m/m). So we have more useful units:

➤ **Parts per Million (ppm)**

- Ratio (by mass) of how many parts of **solute** there are per every **million** parts of the **solution**

➤ **Parts per Billion (ppb)**

- Ratio (by mass) of how many parts of **solute** there are per every **billion** parts of the **solution**

Explore - Calculating ppm & ppb

Parts per Million (ppm)

$$\text{ppm} = \frac{\text{mass solute}}{\text{mass solution}} \times 10^6$$

Masses must be in the
SAME UNIT (Both in
grams is easiest)!!

*These make the number
easy to interpret*

Parts per Billion (ppb)

$$\text{ppb} = \frac{\text{mass solute}}{\text{mass solution}} \times 10^9$$

Explore - Real Life: Lead in Water

The **World Health Organization** says ~**100 ppb** lead in human blood is toxic

- It can cause cancer, and developmental delays in children

Canada regulation - max **5 ppb** for lead in drinking water

US regulation - max **15 ppb** for lead in drinking water

- regulation - max **50 ppb** for lead in river water

This means that safe river water can still have 3x more lead than the limit for safe drinking water

Explore - Calculating ppm & ppb

IN YOUR GROUP - WORK AT A CLASS WHITEBOARD on these two slides
Record your work (take a photo and insert in these slides, redraw it in your notes etc)

1. Health Canada's guideline for maximum Mercury in fish is 0.5ppm. When a 1.6kg salmon was tested, it was found to contain 0.6mg of Mercury. Is this salmon safe to eat? Tip: convert the units to match!

Answer slides at the bottom of slide deck.

Explore - Calculating ppm & ppb

IN YOUR GROUP - WORK AT A CLASS WHITEBOARD

Record your work (take a photo and insert in these slides, redraw it in your notes etc)

2. In Ontario, drinking water should have less than 10 ppb of lead in it. What is the maximum amount of lead allowed in a 500 mL bottle of water?

1mL of water = 1g of water (using density)

Therefore 10ppb = 10g in 1,000,000,000g of water

Set up equivalent Ratios using this data to answer the question!

Answer slides at the bottom of slide deck.

Know your prefixes! See the table below.

The World of Metric Prefixes We Sort Of Care About⁽¹⁾

Now with mostly useless footnotes!

Prefix	Abbreviation	What they mean (human terms)	What they mean (numerically)	What they mean (scientific notation)	An example
mega	M	million	1,000,000	10^6	There are a million bytes i
kilo	k	thousand	1,000	10^3	There are a thousand me
deci	d	tenth	0.1	10^{-1}	There is a tenth of a liter i
centi	c	hundredth	0.01	10^{-2}	There is a hundredth of a
milli	m	thousandth	0.001	10^{-3}	There is a thousandth of a
micro	μ ⁽²⁾	millionth	0.000001	10^{-6}	There is a millionth of a m
nano	n	billionth	0.000000001	10^{-9}	There is a billionth of a m

Concentration Activity

Ask for the Concentration Activity Sheet and visit the stations to complete.

Explore - Molar Concentration

Most useful way to measure concentrations in Chemistry is **moles** per **volume**

➤ Remember it is easy to transfer moles, mass, and particles!

Molar Concentration is the moles of solute per Litre of solution

➤ The units for this are mol/L or **Molarity** (M)

➤ [] is also used for concentration

Ex: $[\text{CuCl}_2] = 2\text{M}$, means the concentration of CuCl_2 is 2 moles per L

$$\text{Molarity (M)} = \frac{\text{moles solute}}{\text{Litre solution}}$$

Explore - Calculating Molar Concentration

Molar Concentration can be calculated as:

The diagram illustrates the formula for calculating molar concentration, $C = \frac{n}{V}$. The variable C is labeled "Concentration" with an orange arrow pointing to it. The variable n is labeled "Moles Solute" with an orange arrow pointing to it. The variable V is labeled "Litres Solution" with an orange arrow pointing to it. The equals sign and the fraction bar are also present.

$$C = \frac{n}{V}$$

Explore - Molarity Practice

A student dissolves 0.212mol of $FeCl_3$ to make a 175mL solution.

a) Calculate the concentration of the solution

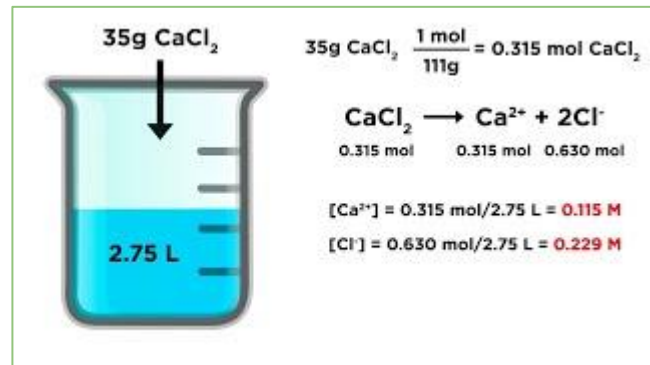
b) Calculate the concentration of chloride ions in solution

a)

$$C = \frac{n}{V} = \frac{0.212 \text{ mol } FeCl_3}{0.175 \text{ L}} = 1.21 \text{ M } FeCl_3$$

b)

$$\begin{array}{l} Fe : Cl \\ 1 : 3 \\ 1.21 : x \\ x = (1.21)(3) = 3.63 \text{ M } Cl^- \end{array}$$



Explore - Mass to Molarity Practice

A student dissolves 29.43g of NaOH into 500mL of distilled water. What is the molar concentration of NaOH in the solution?

Determine Moles:

$$MM_{\text{NaOH}} = 22.99 + 16 + 1.01 = 40$$

$$n = m/MM$$

$$n = 29.43 / 40$$

$$n = 0.73575 \text{ mol}$$

$$500\text{mL} = 0.5 \text{ L}$$

Using moles, determine Concentration:

$$C = n/V$$

$$C = 0.73575 / 0.5$$

$$C = 1.47 \text{ M}$$

$\therefore$ the concentration of NaOH is 1.47M

Explore - Molarity to Mass Practice

***Normal saline solution for rehydration in hospitals contains 0.154M NaCl.
Calculate the mass of NaCl present in 500mL of solution.***

Concentration:

$$500\text{mL} = 0.5 \text{ L}$$

$$C = n/V$$

$$0.154 = n / 0.5\text{L}$$

$$n = 0.077 \text{ mol}$$

Mass:

$$\text{MM}_{\text{NaCl}} = 22.99 + 35.45 = 58.44$$

$$n = m/\text{MM}$$

$$0.077 = m / 58.44$$

$$n = 4.50 \text{ g}$$

$\therefore$ the mass of NaCl in solution is 4.50 g

Explore - Molarity



{Optional} For further help you can watch this video



Part 3: Apply

Apply- Complete the questions



Show all your work! Include formulas, conversions, units, etc.

- 1) (m/v)%: 45g of Cobalt Chloride is dissolved in 231mL of water. What is the (m/V)% Concentration of the solution?
- 2) (m/m)%: Bronze medals in the 2018 PyeongChang Olympics had a mass of 493g. If each bronze medal contained about 435g of Copper, what is the (m/m)% of Copper in the bronze medal?
- 3) (v/v)%: A 300mL bottle of wine contains 35mL of alcohol. What is the (v/v)%
- 4) (v/v)%: A water sample is taken from a river to test for pollution. It must be diluted to 2% by volume of its original amount. If you have a 30mL sample, how much water must you add to get to this desired concentration?
- 5) ppm/ppb: 3mg of a dangerous chemical is found In 300 L of water. What is its concentration in ppm?

Apply- Complete the questions



Work at the board for the next few slides. Insert photos of your work and compare to answers after attempting. Show all your work! Include formulas, conversions, units, etc.

- 1) How many moles of sucrose are dissolved in 250 mL of solution if the solution concentration is 0.150 M?

Apply- Complete the questions



Work at the board for the next few slides. Insert photos of your work and compare to answers after attempting. Show all your work! Include formulas, conversions, units, etc.

2) What is the molarity of a solution of HNO_3 that contains 12.6 grams HNO_3 in 1.0 L of solution?

**Next few slides provide answers to
activities**

Explore - Part B: 5 minutes

Option 1
Moles of solute
per Volume
(liters of solution)

vs

Option 2
Moles of solute
per Weight
(kilograms of solvent)

MOLARITY **VERSUS** **MOLALITY**

Is moles of solute dissolved in 1 liter of solution	Is moles of solute dissolved in 1 kg of solute
Symbol is M	Symbol is m
Formula is mole / liter	Formula is mole / kg
Is affected by changes in temperature	Is not affected by changes in temperature
Is affected by changes in pressure	Is not affected by changes in pressure
May be imprecise or inaccurate	Very precise and accurate

Explore - Calculating ppm & ppb

IN YOUR GROUP - WORK AT A CLASS WHITEBOARD

Record your work (take a photo and insert in these slides, redraw it in your notes etc)

Health Canada's guideline for maximum Mercury in fish is 0.5ppm. When a 1.6kg salmon was tested, it was found to contain 0.6mg of Mercury. Is this salmon safe to eat?

$$\text{ppm} = \frac{0.6\text{mg Mercury}}{1.6\text{kg Fish}} \times 10^6$$

Yes, it is safe to eat.

$$= \frac{0.0006\text{g Mercury}}{1600\text{g Fish}} \times 10^6 = 0.4\text{ppm}$$

Explore - Calculating ppm & ppb

2. In Ontario, drinking water should have less than 10 ppb of lead in it. What is the maximum amount of lead allowed in a 500 mL bottle of water?

1mL of water = 1g of water (using density)

Therefore 10ppb = 10g in 1,000,000,000g of water

⌘ Set up equivalent ratios:

$$\frac{10g}{1 \times 10^9 g} = \frac{x}{500g}$$

$$x = 5 \times 10^{-6} g = 5\mu g$$

Apply- Complete the questions

PLEASE
CHECK
IN

- 1) (m/v)%: 45g of Cobalt Chloride is dissolved in 231mL of water. What is the (m/V)% Concentration of the solution? **Ans: Percent = $m/v \times 100 = 45g/231ml \times 100 = 19\%$**
- 2) (m/m)%: Bronze medals in the 2018 PyeongChang Olympics had a mass of 493g. If each bronze medal contained about 435g of Copper, what is the (m/m)% of Copper in the bronze medal? **Ans: Percent = $m/m \times 100 = 435g/493ml \times 100 = 88.2\%$**
- 3) (v/v)%: A 300mL bottle of wine contains 35mL of alcohol. What is the (v/v)%? **Ans: Percent = $v/v \times 100 = 35ml/300ml \times 100 = 12\%$**
- 4) (v/v)%: A water sample is taken from a river to test for pollution. It must be diluted to 2% by volume of its original amount. If you have a 30mL sample, how much water must you add to get to this desired concentration? **Ans: Percent = $v/v \times 100 \rightarrow 2\% = 30ml/total\ volume \times 100$ Rearrange $\rightarrow Total\ volume = 1500ml$. Amount to add = $total - original\ volume = 1500ml - 30ml = 1470\ ml\ added$**
- 5) ppm/ppb: 3mg of a dangerous chemical is found in 300 L of water. What is its concentration in ppm? **Ppm = $0.003g/300000g \times 10^6 = 0.01\ ppm$**

Apply- Complete the questions



Work at the board for the next few slides. Insert photos of your work and compare to answers after attempting. Show all your work! Include formulas, conversions, units, etc.

- 1) How many moles of sucrose are dissolved in 250 mL of solution if the solution concentration is 0.150 M?

Ans: Convert to L = $250 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.25 \text{ L}$

$$M = n/V$$

$$0.150 \text{ M} = \frac{n}{0.25 \text{ L}}$$

$$n = 0.038 \text{ mol}$$

Apply- Complete the questions

PLEASE
CHECK
IN

Work at the board for the next few slides. Insert photos of your work and compare to answers after attempting. Show all your work! Include formulas, conversions, units, etc.

2) What is the molarity of a solution of HNO_3 that contains 12.6 grams HNO_3 in 1.0 L of solution?

Answer: $n = m / \text{MM}$

$$n \text{ of } \text{HNO}_3 = \frac{12.6 \text{ g } \text{HNO}_3}{63.0 \text{ g } \text{HNO}_3}$$

$$n = 0.200 \text{ mol } \text{HNO}_3$$

$$0.150 \text{ M} = \frac{n}{0.25 \text{ L}}$$

$$M = n/V$$

$$M = \frac{0.200 \text{ mol } \text{HNO}_3}{1.0 \text{ L}}$$

$$M = 0.200 \text{ M}$$

$\therefore$ the molarity of the solution is
0.200 M

Part 5: Beyond the course extensions {optional}